

Enhanced Electrical Properties of Styrene-grafted Polypropylene Insulation for Bulk Power Transmission HVDC Cables

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Abstract—Polypropylene (PP) is considered to be a rather promising material for HVDC cable insulation. Grafting modification has proved to be an effective method on the electrical property improvements of PP. This paper reports on the highly-enhanced electrical properties of styrene-grafted polypropylene (PP-g-St) and their related mechanism. The influence of grafting styrene on crystallization structures, thermal properties and electrical properties of PP-g-St are characterized and analyzed. These results show that PP-g-St exhibits remarkably enhanced DC volume resistivity, suppressed space charge accumulation and improved DC breakdown strengths as compared to PP. Crystallization observation shows PP-g-St has small, imperfect spherulites with indistinct boundaries, which is strongly correlated to enhanced electrical properties. This study provides an effective grafting strategy to design high-performance HVDC cable insulation materials for bulk power transmission systems.

Index Terms—Electrical properties, grafting modification, polypropylene, styrene.

I. INTRODUCTION

POWER cables are an important part of the power transmission system. Along with the accelerated urbanization and the development of renewable resources, the demand for power cables is continuously growing. Particularly, the development of flexible DC transmission drives the rapid development of high voltage direct current (HVDC) cables [1]–[5]. Currently, cross-linked polyethylene (XLPE) is extensively used as the main insulation material in HVDC cables [6]–[11]. However, the operating temperature of XLPE-insulated HVDC cable is limited within 70°C, which cannot meet the demand of the further rising capacity of transmission systems [12],

[13]. In addition, XLPE is thermoset, thus unable to be recycled after the cable decommissioning. The traditional process methods, such as incineration, cracking and burial, not only consume a large amount of energy, but also have a severely negative impact on the environment [14]. As for their production, the process of polyethylene crosslinking requires an inert environment of high pressure and high temperature, and the extrusion process also needs to be precisely controlled to prevent the phenomenon of scorching, resulting in a relatively complicated production process. In addition, the cross-linking by-product is easy to cause microporous defects and partial discharges, which could easily trigger severe space charge accumulation under high DC electric field, eventually seriously shortening the service life of the cable [13], [15]–[17]. In view of the above shortcomings, XLPE has been unable to meet the needs of green and sustainable contemporary social and economic development, and it is necessary to develop a new insulation material for HVDC cables to replace XLPE [18].

As a thermoplastic polymer, polypropylene (PP) shows excellent insulation properties, high temperature tolerance, chemical corrosion resistance, plastic recyclable nature, etc. These advantages, which makes PP superior to XLPE, shows that PP could act as the next generation HVDC cable insulation material [19]–[21]. To meet the operating demands under higher electric fields and temperatures, it is important to further enhance the insulation properties of PP. At present, studies on PP electrical properties modification primarily focus on two aspects: nano-composition and grafting modification. Montanari *et al.* discovered that adding montmorillonite (MMT) nanomaterials into PP can enhance the DC breakdown strength and DC volume resistivity of nanocomposites [22]. Zhou *et al.* studied the effects of different inorganic nanoparticles, such as MgO [23]–[25], ZnO [26], TiO₂ [27] and fullerene [28] and different nanoparticle surface-modification methods on the electrical properties of PP-based nanocomposites [17], finding that the nanocomposite of modified MgO nanoparticles with isotropic polypropylene (iPP) shows the best improvement in dielectric performance.

Although decent enhancements of the electrical properties of PP have been achieved by nano-composition, it is strongly dependent on the nanoparticle dispersion in the PP matrix. Actually, it is difficult for nanoparticles to be uniformly dispersed. Nanoparticle aggregation is easy to occur, which will

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deteriorate the electrical properties of the nanocomposite [29]. So the properties stability of products is difficult to control, which limits their practical applications [30]. In comparison, chemical grafting modification, which chemically modifies the polymer macromolecule, is also an effective method to enhance the electrical properties of PP, meanwhile avoiding the nanoparticle aggregation, and it has become one of the research hotspots. Zha *et al.* prepared maleic anhydride-grafted PP by the reactive extrusion method, and found that the average size of PP spherulite decreases with the grafting content of maleic anhydride increasing, and the space charge injection and accumulation were suppressed [31]. Hai *et al.* grafted poly (3-hexylthiophene) onto PP macromolecule, and found that the thermal stability and electrical resistivity of PP were improved [32]. Thus, it can be accepted that the grafting modification of PP has become an effective and available method to enhance the electrical properties of PP.

Zhou *et al.* grafted styrene onto the PP macromolecule, and found that the grafting styrene group can also improve the electrical properties of PP. They explained its mechanism of deep trap introduction by the thermally stimulated current (TSC) test and density functional theory (DFT) simulation [33]. However, the grafting content is also an important factor that affects the modification effect and needs to be further studied. In this study, iPP is chosen as the matrix, and styrene with different grafting content is grafted onto PP by the water phase suspension grafting method. The effect of styrene content on the thermal properties, crystallization structure and electrical properties of prepared grafting modified PP (PP-g-St) were analyzed and discussed. This study aims to be a reference for mechanism research on grafting modification of PP and to provide an effective modification strategy for HVDC cable insulation material.

II. EXPERIMENT

A. Materials

Polypropylene was provided by the Maoming Petrochemical Company with a density of 0.9 g/cm³. Grafting monomer styrene (St) and initiator benzoyl peroxide (BPO) were purchased from the Beijing J&K Technology Co. Ltd. Ethyl acetate was produced by the Sinopharm Chemical Reagent Co. Ltd. Styrene was purified by vacuum distillation before use to remove any polymerization inhibitors. Other reagents were used without treatment.

B. Preparation of Styrene-grafted PP

The PP matrix was dried in a vacuum oven at 60°C for 6 h, and then was added into the reactor by mechanical stirring. Nitrogen purge was used to remove oxygen in the system. Under the protection of nitrogen, the prepared St-BPO solution was added into the reaction kettle, and stirred and evenly mixed. The mixture was first heated to 60°C for 2 h, then further heated to 90°C with 30 min stirring. Then, deionized water was added into the reactor for further reaction for 4 h. After that, the mixture went through the process of filtration, nitrogen purge and drying, and then was added into a granulator to obtain the granules. The styrene-grafted PP

was eventually prepared. During the preparation, the styrene content (grafting content) was controlled as 2.5%, 5%, 10%, and 20% of the PP matrix (weight percent), named as PP-g-St2.5, PP-g-St5, PP-g-St10, PP-g-St20, respectively.

To calculate the grafting rate (GR) and grafting efficiency (GE), the grafted PP was added into a Soxhlet extractor, and then extracted with ethyl acetate for 24 h to remove unreacted and self-polymerized styrene. After that, the grafted PP was dried and weighed. GR and GE can be calculated as (1) and (2):

$$GR = \frac{w_1 - w_0}{w_1} \times 100\% \quad (1)$$

$$GE = \frac{w_1 - w_0}{w_2} \times 100\% \quad (2)$$

where w_1 is the weight of grafted PP after extraction, w_0 is the weight of PP before grafting modification, w_2 is the weight of the added styrene [34].

To prepare the thin film samples for the following characterization, the styrene-grafted PP and pure PP granules were compression molded at 200°C, 15 MPa for 10 min, and then hydro-cooled to room temperature (RT, about 25°C) at 10 MPa for 5 min.

C. Characterization

The Fourier-transform infrared spectrometer (FT-IR, Thermo Fisher Nicolet iS10, USA) was used to characterize the grafting chemical group of styrene. The resolution was 4 cm⁻¹. The melt mass-flow rate (MFR) was measured by a MFR meter (CEAST7026, Italy) at 230°C, with the experimental load of 2.16 kg. To study the thermal properties, a differential scanning calorimetry (DSC, Perkin-Elmer Diamond DSC, USA) was carried out. Under the protection of nitrogen, the sample was first heated to 200°C at the rate of 10°C/min for 3 min to erase heat history. Then it was cooled to 50°C and heated to 200°C again to measure the melting process. The melting enthalpy during the melting process was measured to calculate the crystallinity of pure and styrene-grafted PP, where the melting enthalpy of 100% crystalline PP is 209 J/g.

To investigate the crystallization of styrene-grafted PP, X-ray diffraction (XRD, Philips X' Pert MPD, Netherlands) and polarizing microscopy (POM, Zeiss Axio Imager A1m, Germany) were used to characterize the crystal form and observe the spherulite morphology, respectively. The 2θ angle scanning range of XRD was 5° to 35°, with the scanning rate of 3°/min. During POM observation, the samples were heated to 210°C for 3 min to erase heat history, then cooled to RT at 10°C/min, meanwhile the POM images were captured.

DC volume resistivity was calculated from the DC leakage current, which was tested by a three-electrode platform and measured by an electrometer (Keithley 2635B, USA) at RT, 50, 70, and 90°C. The film sample of 100 μm average thickness was sputtering-coated with gold electrodes. The applied electric field ranges were from 10 kV/mm to 80 kV/mm, and it rose by 5 kV/mm for each measurement period. The pulsed electroacoustic method (PEA) was adopted to measure the space charge behavior of samples under RT. The average sample

thickness was 200 μm . A 40 kV/mm DC electric field was applied to the samples for 30 min and then short-circuited for 10 min. The charge density was calculated from the electro-acoustic waveform. A DC high voltage tester (Haiwo Science & Technology, China) was employed to test the DC electric breakdown strength. The sample whose thickness was about 100 μm was sandwiched by sphere-plate electrodes. A 1 kV/s increasing voltage was applied between the electrodes until breakdown. The test platform was soaked in silicone oil to avoid surface flashover. 20 different points were tested for each sample, meanwhile the breakdown voltage values (U) were recorded. The breakdown strength E was calculated by $E = U/d$ for each point, where d is the thickness of the point. The breakdown strength data were analyzed by two-parameter Weibull distribution, as described in Equation (3):

$$P(E) = 1 - \exp(-(E/E_0)^\beta) \quad (3)$$

where $P(E)$ is the accumulative breakdown probability. E is the aforementioned breakdown strength. E_0 is the characteristic breakdown strength at the accumulative breakdown probability of $1 - e^{-1}$ (approximately 63.2%). β represents the shape parameter of the Weibull fitting. The DC breakdown strength test was performed under RT, 50, 70, and 90°C.

III. RESULTS AND DISCUSSION

A. Characterization of Styrene-grafted PP

Table I shows the composition of pure PP and PP-g-St. One can see that not all the styrene monomers are grafted onto PP. With the styrene content increasing, the grafting efficiency rises first, reaching the peak in PP-g-St5, and then decreases. After the grafting process, there may be some styrene and BPO still remaining unreacted, and some styrene could self-polymerize. The raised styrene content can result in a higher grafting reaction rate, thus increasing the grafting efficiency. However, the styrene self-polymerization rate at high content can also rise, which leads to reduced efficiency.

TABLE I
COMPOSITION OF PP AND PP-G-ST

Sample	Styrene Content (%)	Weight Gain Rate (%)	Grafting Rate (%)	Grafting Efficiency (%)
Pure PP	0	0	0	0
PP-g-St2.5	2.5	2.3	1.4	55.6
PP-g-St5	5	4.9	3.0	62.9
PP-g-St10	10	9.8	5.5	58.4
PP-g-St20	20	19.8	8.5	46.4

The FT-IR spectra of PP-g-St with different grafting content is shown in Fig. 1. The absorption peaks at the wave number of 1584 cm^{-1} and 1602 cm^{-1} which corresponds to the stretch vibration of the aromatic ring skeleton [34]. And the peak value is nearly proportional to the grafting content. Thus, it could be confirmed that the styrene is successfully grafted onto PP.

B. Thermal Properties of Styrene-grafted PP

Figure 2 shows the crystallization curves (Fig. 2(a)) and melting curves (Fig. 2(b)) of pure PP and PP-g-St by DSC.

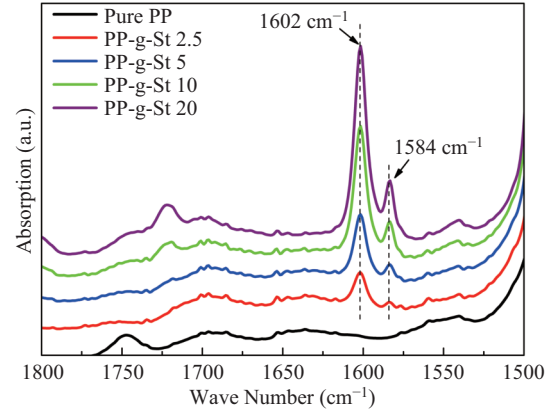


Fig. 1. FT-IR spectra of pure PP and PP-g-St with different grafting content.

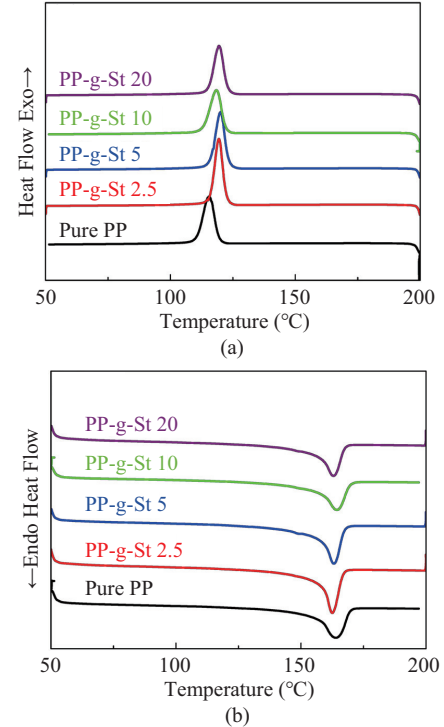


Fig. 2. DSC curves of PP and PP-g-St with different grafting content. (a) Crystallization curve. (b) Melting curve.

Table II exhibits the thermal parameters measured by DSC and MFR. The melting temperature barely changes, but the crystallization temperature of PP-g-St is obviously higher than that of pure PP, and along with crystallinity, they first increase then decrease with the styrene content increasing. These results indicate that the grafted styrene group can promote the crystallization of PP. Compared with PP, the grafted groups, branches, and self-polymerized styrene show different shrinkage coefficients during the cooling process, so that they would generate internal stress against PP, then promoting nucleation and crystallization. However, when the styrene content further rises, due to the steric effect of the aromatic ring group, the grafted PP molecular chain would be more difficult to crystallize and to shape into perfect spherulites. Thus, the crystallinity decreased. In addition, the steric effect can also restrict the relative motion between PP

TABLE II
THERMAL PARAMETERS OF PP AND PP-G-ST

Sample	Melting Temp. (°C)	Crystallization Temp. (°C)	Melting Enthalpy (J/g)	Crystallinity (%)	Melt Massflow Rate (g/10 min)
Pure PP	163.87	115.47	96.54	46.64	3.9
PP-g-St2.5	162.50	119.44	98.55	47.61	2.2
PP-g-St5	163.16	120.05	101.20	48.89	2.0
PP-g-St10	164.29	118.44	90.94	43.93	1.5
PP-g-St20	162.99	119.49	90.98	43.95	1.4

molecular chains. This is a possible reason why the MFR of PP-g-St is significantly lower than that of pure PP.

C. Crystallization Characteristics

Figure 3 shows the XRD spectra of pure PP and PP-g-St. 5 main diffraction peaks at 14.1° , 16.9° , 18.5° , 21.1° , and 21.9° appear in each sample. The corresponding crystal faces of these 2θ angles are (110), (040), (130), (111), and (131), belonging to the alpha crystal forms of PP [35], [36]. The XRD result indicates that both pure PP and PP-g-St are of alpha crystal form, and grafting styrene has no obvious effect

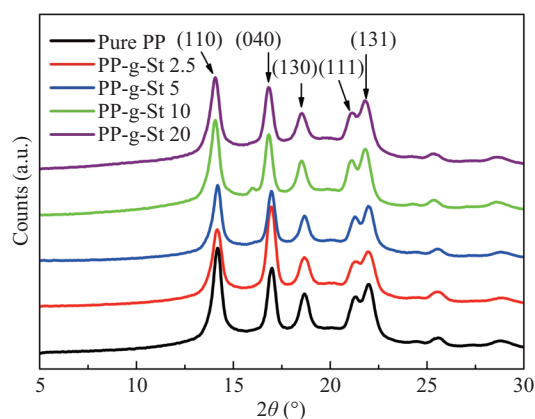


Fig. 3. XRD spectra of PP and PP-g-St with different grafting content.

on the crystal form of PP.

Figure 4 exhibits the POM images of pure PP and PP-g-St. It can be seen that perfect spherulites with clear boundaries are shaped in pure PP. After grafting styrene, the spherulite size reduces in PP-g-St. Among them, PP-g-St5 shows the smallest spherulites. With the styrene content rising, the spherulite boundaries become more indistinct, and the regularity of spherulite declines: e.g., the obvious imperfect spherulites can be observed in PP-g-St5. In the POM images at high grafting content, such as PP-g-St10 and PP-g-St20, quantities of black spots can be observed and are believed to be the impurities (self-polymerized styrene, unreacted styrene and BPO) introduced during the grafting modification process.

During the cooling process, as described in Section III-B, the side groups, branches and self-polymerized styrene can promote the nucleation. Thus, the quantity of spherulite rises, and the size decreases. In addition, the grafted styrene group would destroy the regularity of the PP molecular chain, so that the molecular motions of PP, such as folding, and arrangement, are hindered. Finally, the regularity of the spherulites worsens, and the boundaries are weakened. As for high styrene content situations, i.e., PP-g-St10 and PP-g-St20, more styrene would self-polymerize and be expelled from the crystalline region, therefore weakening the nucleation effect of styrene. Therefore, the spherulite is larger and more regular than that of PP-g-St5.

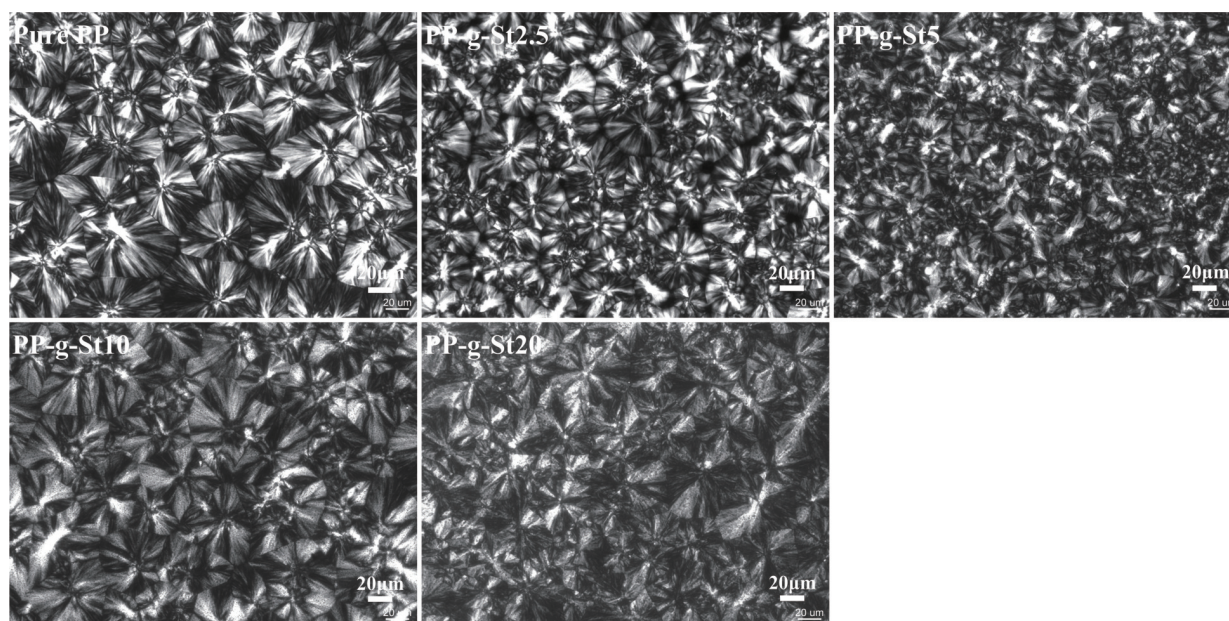


Fig. 4. POM images of PP and PP-g-St with different grafting content.

D. DC Volume Resistivity

DC volume resistivity calculated from the leakage current is an essential electrical property index of HVDC cable insulation. Fig. 5 shows the resistivity values of pure PP and PP-g-St with different grafting content under RT, 50, 70 and 90°C.

As shown in Fig. 5, the DC volume resistivity of styrene-grafted PP is significantly raised compared with pure PP, especially under high temperatures and high electric fields. This indicates that styrene-grafted PP shows lower leakage current, which means lower charge carrier mobility.

Some related research points out that the low leakage current of grafting modified PP is strongly correlated to the large quantities of deep traps introduced by the grafted group [33], [37]. The grafted aromatic ring group can effectively change the electrical potential distribution of the PP molecular chain, and introduce the trap energy level within the band gap. Deep traps could capture charge carriers and reduce the charge mobility, resulting in low leakage current and high DC resistivity [33]. Under high temperature, the motion of the PP molecular chain is very fierce, and the charge mobility is high, resulting in high leakage current and low resistivity, meanwhile the effect of the electric field is low. However, as for PP-g-St, the capturing effect of deep traps is more significant than that under RT, so that PP-g-St shows more obvious improvements of resistivity.

Comparing the various styrene-grafted PP, the resistivity of PP-g-St5/10/20 is higher than that of PP-g-St2.5 especially under a high electric field. This could be explained as that high grafting content means a more grafted aromatic ring group and more introduced deep traps. In addition, the amorphous region existing in PP usually shows relatively low resistivity, and is easier to conduct charge carriers. PP with high grafting content has more small, imperfect spherulites with fuzzy boundaries. The highly interlaced amorphous region and crystalline region make the conduction paths longer and more incomplete. Thus, the charge mobility reduces, eventually leading to high resistivity.

E. Space Charge Distribution

Space charge behavior is a rather important issue for HVDC cable insulation. Space charge accumulation would induce electric field distortion, triggering partial discharge and other severe faults within the cable. Fig. 6 exhibits the dynamic space charge distributions of PP and PP-g-St under RT. Fig. 6(a)–(e) correspond to the accumulation process within 30 min, while (f)–(j) are of the space charge dispersion during a 10 min short-circuit.

As Fig. 6(a) shows, pure PP exhibits obvious homocharge accumulation near the cathode. After the styrene group is grafted, the homocharge accumulation near the cathode is suppressed. Especially in PP-g-St2.5, as Fig. 6(b) shows, the

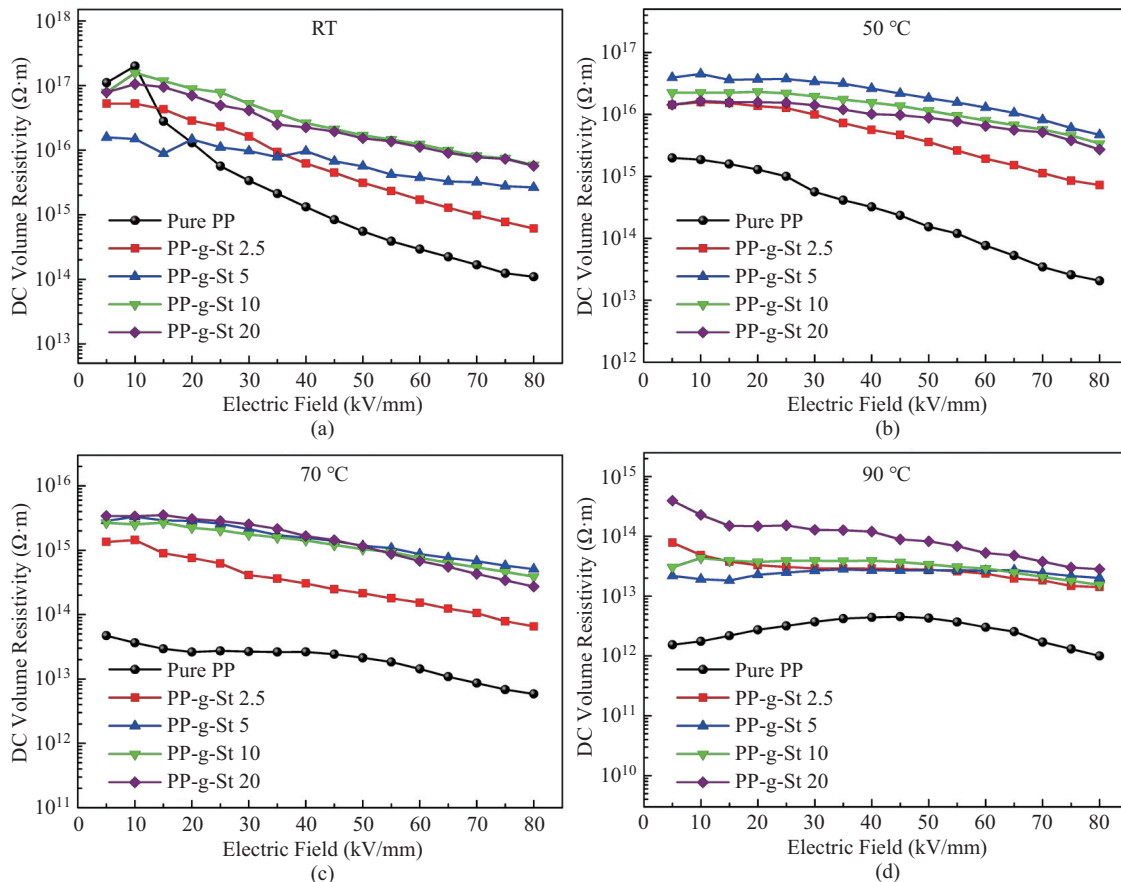


Fig. 5. DC volume resistivity of pure PP and PP-g-St under the DC electric field ranging from 5 to 80 kV/mm and different temperature. (a) 30°C. (b) 50°C. (c) 70°C. (d) 90°C.

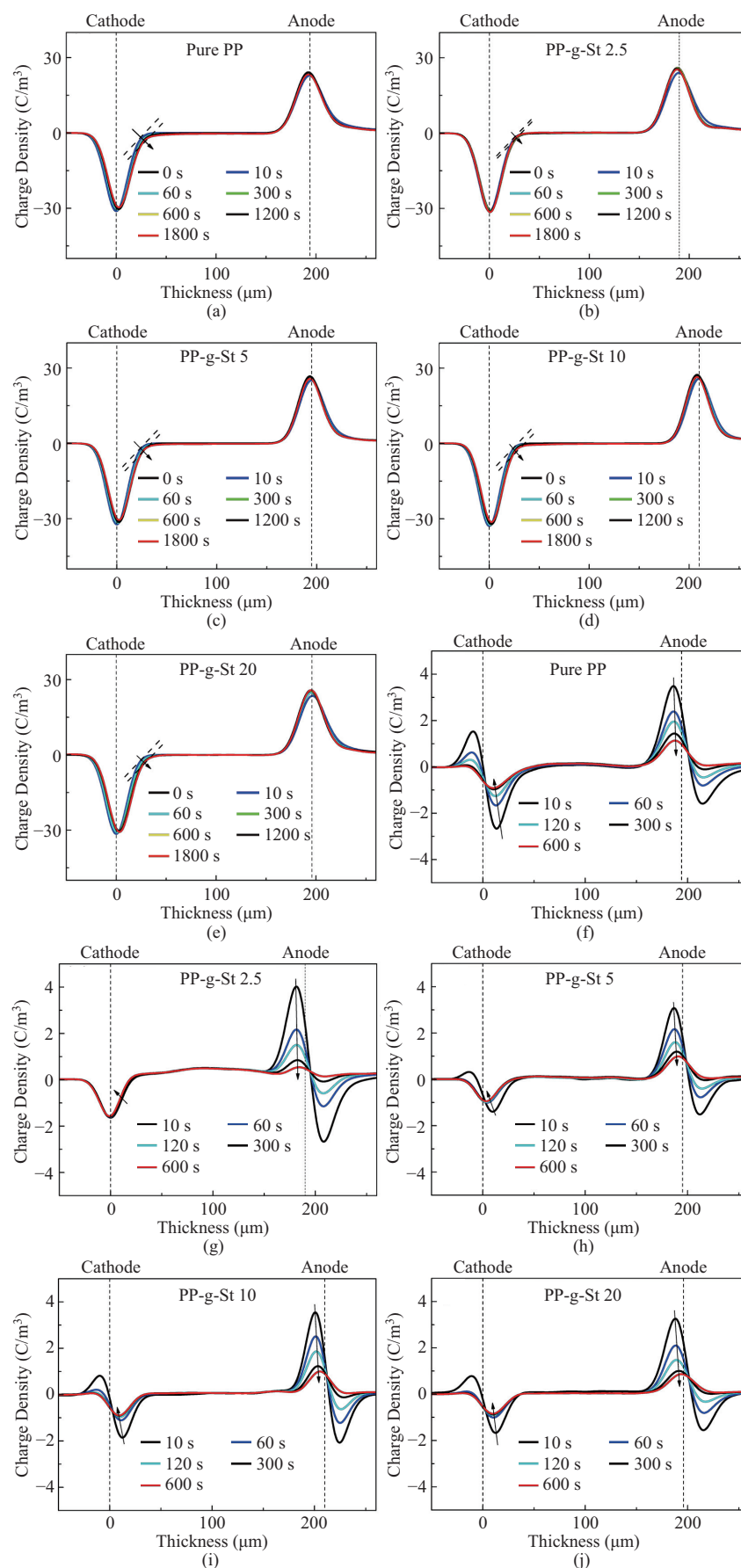


Fig. 6. Dynamic space charge distribution of pure PP and PP-g-St with different grafting content. (a)–(e) Space charge accumulation. (f)–(j) Space charge dispersion.

homocharge injection can hardly be seen. However, with the grafting content increasing, the homocharge accumulation is visible again. The space charge suppression in PP-g-St can be attributed to the deep traps introduced by the grafted styrene group. Deep traps could capture injected charge carriers. The captured charges near the cathode would weaken the local electric field, so that the follow-up injection of homocharges is suppressed [38], [39].

When the grafting content is further raised, as demonstrated in Section III.C, the spherulites are irregular, imperfect, with more indistinct boundaries. This would introduce quantities of defects. In addition, the impurities introduced by the

grafting process, such as unreacted styrene and BPO, and self-polymerized styrene, could also act as defects in the amorphous region. These defects in the amorphous region and boundaries are the weak points in the polymer, and would trigger space charge accumulation under a high electric field, so that the space charge accumulation in PP of the high grafting content deteriorates.

The residual charges and dispersion process can be seen in Fig. 6(f) to (j). Compared with pure PP, the initial residual charges near the cathode in PP-g-St are less, and disperse rapidly at first, then slow down. These results indicate that the injected charges near the cathode in PP-g-St are less than pure

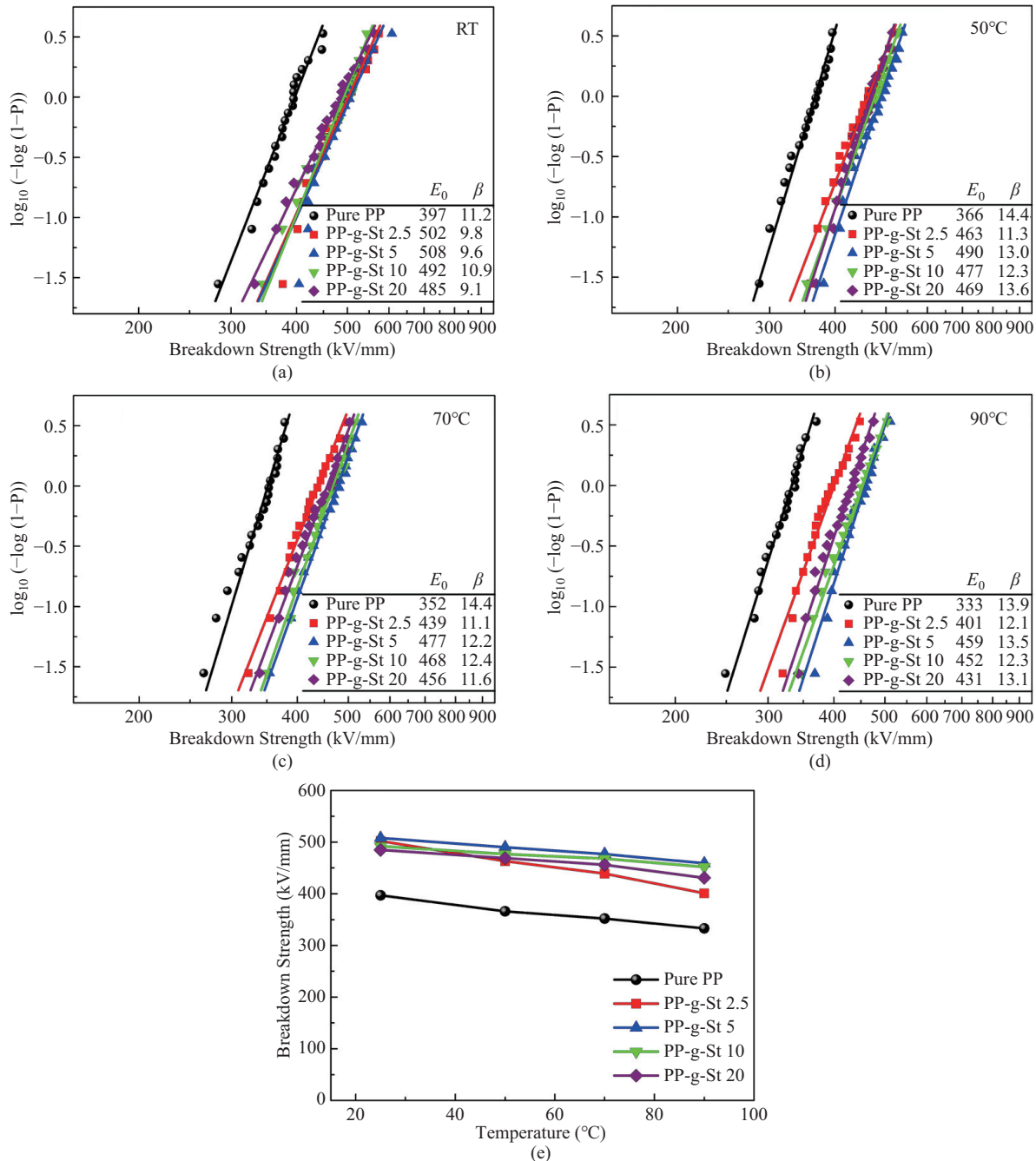


Fig. 7. DC breakdown strength Weibull fitting plots of PP and PP-g-St under (a) room temperature, (b) 50°C, (c) 70°C, (d) 90°C, and (e) the temperature dependent DC breakdown strength values.

PP. Due to the deep traps introduced by the grafted styrene group, the captured space charges are difficult to disperse. So that the residual charge density from 60 s to 600 s slightly changes.

F. DC Breakdown Strength

Bulk power transmission usually means high voltage levels and heavy load currents. It demands that the HVDC cable insulation material must possess high DC breakdown strength to tolerate high voltages, and excellent resistance against high temperatures originating from heavy current. Therefore, the DC breakdown strength under high temperatures is rather important for insulation material. Fig. 7 exhibits the DC breakdown strength Weibull fitting plots of pure PP and PP-g-St under different temperatures (Fig. 7(a) to (d)) and the temperature dependent DC breakdown strength values (Fig. 7(e)).

As Fig. 7 shows, compared with pure PP, the DC breakdown strength values of PP-g-St significantly rise under all test temperatures. With the grafting content increasing, the breakdown strength first rises and then slightly decreases. Among all samples, PP-g-St5 shows the highest breakdown strength, which rises by 28% at RT, 34%, 36%, 38% at 50, 70, 90°C respectively (compared with pure PP). These results indicate that styrene-grafted PP can effectively enhance the DC high voltage tolerance under high temperatures.

Some research indicates that the electrical breakdown process in semi-crystalline polymer is strongly correlated to crystalline boundaries [40]–[43]. In the case of PP, Spherulite boundaries show relatively low charge carrier resistivity, and would be more likely to be a paved electrical breakdown path. After grafting styrene, PP spherulites show reduced size, indistinct boundaries, and imperfect crystalline forms. This would make the carrier transmission path longer and destroy the integrity of the charge conduction route, making the DC breakdown strength enhanced. In addition, the deep traps introduced by grafted styrene can also capture high-energy charge carriers during the breakdown process, reducing their damage to the PP molecular chain, eventually increasing the breakdown strength.

The grafting content dependent breakdown strength can be explained as follows. In the situation of low content, the grafting effect on crystallization of PP is limited. The introduced deep trap quantity is also low. Therefore, the breakdown strength enhancement of low-content grafted PP is restricted. At high grafting content, as stated in Section III.C, more self-polymerized styrene would be expelled from the crystalline region, and act as impurities. Collaborating with other impurities introduced by the grafting process, such as unreacted styrene and BPO, they may introduce defects in the material, deteriorating the breakdown performance. Therefore, the breakdown strength of high-content grafted PP decreases.

IV. CONCLUSION

Styrene-grafted polypropylene (PP-g-St) is a possible HVDC cable insulation material to replace XLPE in bulk power transmissions. The thermal properties, crystallization

characteristics and electrical properties under high temperatures of PP-g-St were tested. The effect of grafting content and the related mechanism were studied. The experimental results and conclusions are as follows:

1) When grafting content rises, the melting temperature slightly changes, whereas the melt mass-flow rate reduces, which could be attributed to the steric effect of the grafted styrene group restricting the motion of the PP molecular chain.

2) XRD characterization shows that both PP and PP-g-St are of alpha crystal form. With the grafting content rising, the crystallinity first rises and then decreases, whereas the spherulite size shows the opposite tendency. The spherulite boundaries of PP-g-St are indistinct, and the crystalline form is imperfect compared with pure PP.

3) The electrical properties of PP-g-St are significantly enhanced, especially under high temperatures. Compared with pure PP, the DC volume resistivity and DC breakdown strength are enhanced by 2 orders of magnitude and 38%, respectively. The space charge accumulation near the cathode is also suppressed. These enhanced electrical properties can be attributed to the fact that grafted styrene affects the crystallization of PP, resulting in small spherulite sizes and indistinct boundaries, then weakening and destroying the charge transmission path. However, excessively high grafting content will introduce impurities in the PP matrix and deteriorate the electrical properties. This study provides a reference on the strategy and application of grafting modification on designing HVDC cable insulation material for bulk power transmission systems.

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